Quiz 4 - DSC 10, Winter 2026

Full Name:

PID:

Quiz Time: ◯ 1PM ◯ 2PM

Instructions:

- This quiz consists of 6 questions. You have a total of 30 minutes to complete it.
- Please write **clearly** in the provided answer boxes; we will not grade work that appears elsewhere. Completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.

  ◯ A bubble means that you should only **select one choice**.

  ☐ A square box means you should **select all that apply**.
- If your answer is a string, make sure to put it in quotes. If your answer is a float, make sure to include a decimal point.
- You may use one handwritten sheet of notes. No calculators and no computers.
- Assume we have already run `import babypandas as bpd` and `import numpy as np`.

Person to your left:

Person to your right:

By signing below, you are agreeing that you will behave honestly and fairly during and after this quiz.

Signature:

Version A

Please do not open your quiz until instructed to do so.

We are using (fictional) grading data from Quiz 3. Jeffrey and Kate each graded a random sample of DSC 10 submissions. The `grading` DataFrame has one row per graded submission with 50 rows total (25 per grader). The columns are: `"grader"` (str, either `"Jeffrey"` or `"Kate"`), `"score"` (int, score between 0 and 100), and `"submission_id"` (int, unique ID). The first few rows of `grading` are shown below.

	grader	score	submission_id
0	Jeffrey	65	1
1	Kate	78	2
2	Jeffrey	70	3
3	Kate	82	4
4	Jeffrey	68	5
		...	

After examining the difficulty of the last DSC 10 quiz, expert detective and professor Peter has deduced that one of the two graders has been grading significantly harsher than the other.

Question 1: Setting a Trap!

Using his years of statistical skill and investigative prowess, Detective Peter decides to conduct a permutation test to determine whether one of the two graders was significantly harsher than the other. He has Jeffrey's scores and Kate's scores and no assumed knowledge of the population distribution of scores. Which of the following best explains both **why** a permutation test is appropriate here and **how** it should be done?

◯ We have two samples and want to test whether they came from the same distribution. Under the null hypothesis, we simulate by shuffling the group labels, since if the distributions are the same the labels shouldn't matter.

◯ We have two samples and want to test whether they came from the same distribution. Under the null hypothesis, we simulate by resampling with replacement from the combined 50 scores to form two new groups.

◯ We want to compare Jeffrey's scores against a known population distribution of Quiz 3 scores. Under the null hypothesis, we simulate by shuffling scores and recomputing the test statistic.

◯ Shuffling the grader labels generates two new random samples from the population of all possible Quiz 3 scores, letting us estimate the sampling distribution of the test statistic.

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Question 2: Building the Case

Detective Peter moves forward with his permutation test to compare Jeffrey's and Kate's scores. He suspects that Jeffrey is the harsher grader — that is, Jeffrey's scores tend to be lower than Kate's, and wants to determine if there is evidence to support this.

a) Which **null and alternative hypothesis pair** is most appropriate?

◯ **Null:** Jeffrey and Kate gave scores from the same distribution. **Alt:** Jeffrey gave scores from a distribution with a smaller mean than Kate's.

◯ **Null:** Jeffrey's scores come from a known population with mean 70. **Alt:** Jeffrey's scores do not come from a population with mean 70.

◯ **Null:** Jeffrey and Kate gave scores from the same distribution. **Alt:** The distribution of Jeffrey's scores is entirely lower than the distribution of Kate's scores.

◯ **Null:** The mean score from Jeffrey equals the mean score from Kate. **Alt:** The mean score from Jeffrey does not equal the mean score from Kate.

b) For testing whether Jeffrey is the harsher grader, which of the following could be used as a test statistic to get a p-value directly from its empirical distribution under the null hypothesis? Select all that are correct.

☐ The difference in group means: Kate's mean minus Jeffrey's mean.
☐ The difference in group means: Jeffrey's mean minus Kate's mean.
☐ The absolute difference between the two groups' mean scores.
☐ The total variation distance between the two graders' score distributions.
☐ None of the above.

Question 3: Running the Numbers

Detective Peter asks another staff member, Ray, to write code to run the permutation test with 10,000 permutations. Ray comes up with his own test statistic and writes the following code to calculate an observed statistic, stored in `observed_stat`. Ray also wrote code (not shown) to store 10,000 simulated test statistics, each calculated in a manner matching the code for the observed statistic, in the array called `simulated_stats`. Fill in the two blanks below so it correctly computes the p-value for testing whether Jeffrey is the harsher grader.

```
observed_stat = (abs(grading[grading.get('grader') == 'Kate']
    .get('score').mean())
    - abs(grading[grading.get('grader') == 'Jeffrey']
    .get('score').mean()))

p_value = (simulated_stats __(a)__ observed_stat).sum() / __(b)__
```

(a): _____ (b): _____

Question 4: Understanding the Damage

After finding out through his expertly constructed permutation test that Jeffrey did, in fact, grade significantly harder than Kate, Detective Peter resolves to estimate the **proportion** of Jeffrey's graded submissions that received a score below 70. He will construct a **95% CLT-based confidence interval** for this proportion.

a) He wants the confidence interval to have **width at most 0.08**. What is the **minimum** number of Jeffrey's submissions he needs?

b) Suppose he instead wants the width to be **at most 0.04**. The minimum sample size for that width is how many **times** the minimum sample size from part (a)?

Question 5: Jeffrey's Alibi

Jeffrey frantically defends himself by claiming that the true proportion of his submissions that scored below 70 is really **0.20** — not nearly as bad as it looks! To test his claim, Detective Peter constructs a 95% confidence interval and gets **[0.14, 0.40]**.

a) Using this confidence interval, at the **5% significance level**, what decision should we make with respect to Jeffrey's claim? Select all that are correct.

☐ Reject ☐ Fail to reject ☐ Accept ☐ Not enough information

b) Which of the following best explains why we reach that conclusion?

◯ Because 0.20 lies inside the confidence interval, the data are consistent with Jeffrey's claim.

◯ Because 0.20 lies inside the confidence interval, the data are inconsistent with Jeffrey's claim.

◯ Because the interval has a width of 0.26 which is much wider than we wanted, we do **not** have enough precision to reject any claim at the 5% significance level.

◯ Because the midpoint of the interval is 0.27, we have evidence against Jeffrey's claim.

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Question 6: Jeffrey's Penance

Overall, Detective Peter determined that Jeffrey's offenses are grave enough: Peter travels all the way back to Tutorlandia to consult the great council of tutors on what Jeffrey's penance should be. It was decided that Jeffrey must explain to the class what the **total variation distance (TVD)** measures. Which of the following is a correct description of the TVD?

◯ The difference between two group means.
◯ A measure of how different two distributions (of any type) are.
◯ A measure of how different two **numeric** distributions are.
◯ A measure of how different two **categorical** distributions are.
◯ The proportion of simulated statistics that are at least as extreme as the observed value.
◯ The width of a 95% confidence interval.

Reminders:

- **Write your PID** on the front page and on the top right corner of page 3.
- If your answer is a string, make sure to put it in **quotes**.
- If your answer is a float, make sure to include a **decimal point**.
- Stay in your seat until the quiz is over. No leaving early.